

Predicting Qualifying Scores in NCAA Diving Competition

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Project Overview

- Women's NCAA Diving Scores Analysis
- Prequalification scores determine whether each diver qualifies for competition
- Analyze patterns within the score data to predict qualifying score for upcoming diving season
- Assist coaches and divers by providing qualification insights for improved training methods



Data Processing

- Read in data from “womens_individual.csv”
- Create separate datasets for 1 meter and 3 meter events by filtering
- Create variable, “highest_prequal”, which takes the higher score for each diver’s 1 meter or 3 meter event
- Create separate datasets for qualifying and non-qualifying divers to explore their differences
- One observation removed from dataset (3 meter prequalification score of 365.5 for a single athlete in 2012) due to an implausible score and most likely a data entry error.

EDA

Exploring Potential Predictors: Class Rank



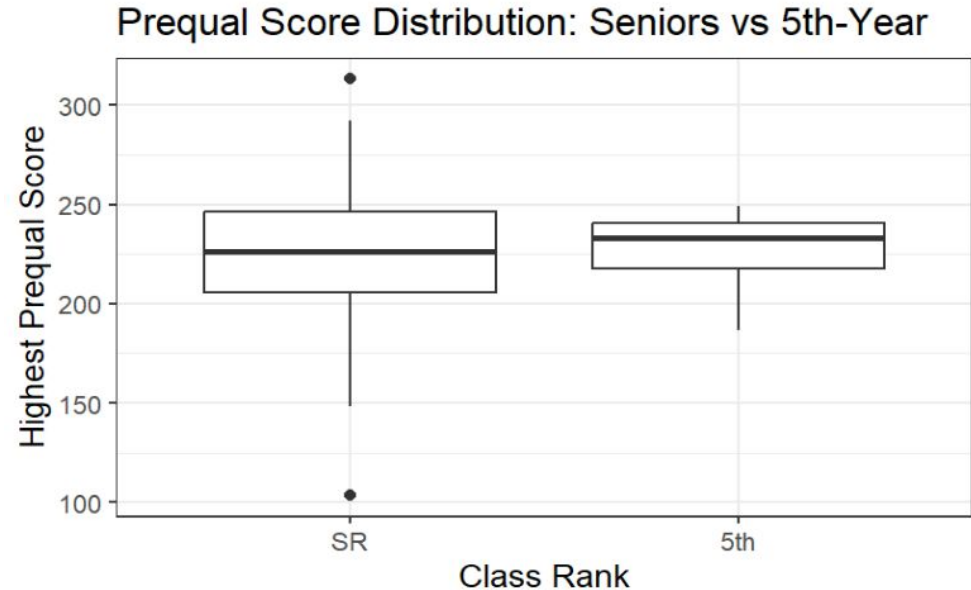
5th years and seniors have higher proportion of qualification.

Freshmen have the lowest proportion of qualification.

Due to small sample, **should 5th years be grouped with seniors?**

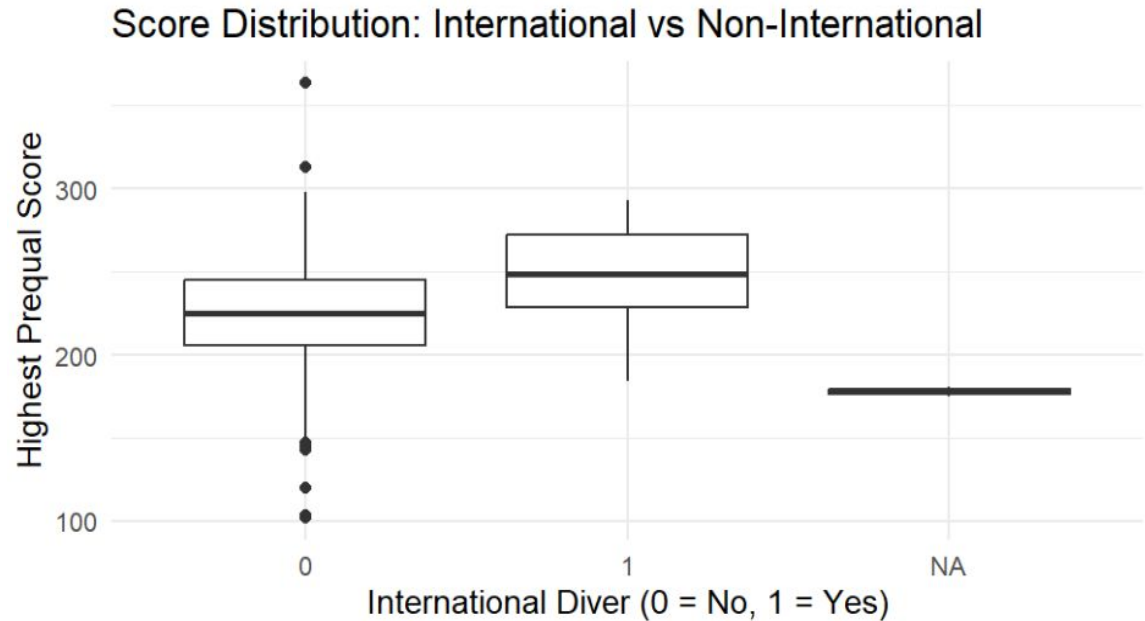
How should we deal with 5th years?

- Because the means between these two groups are so close and there are only 6 data points for 5th year divers.
- We decide to **merge Seniors and 5th years.**

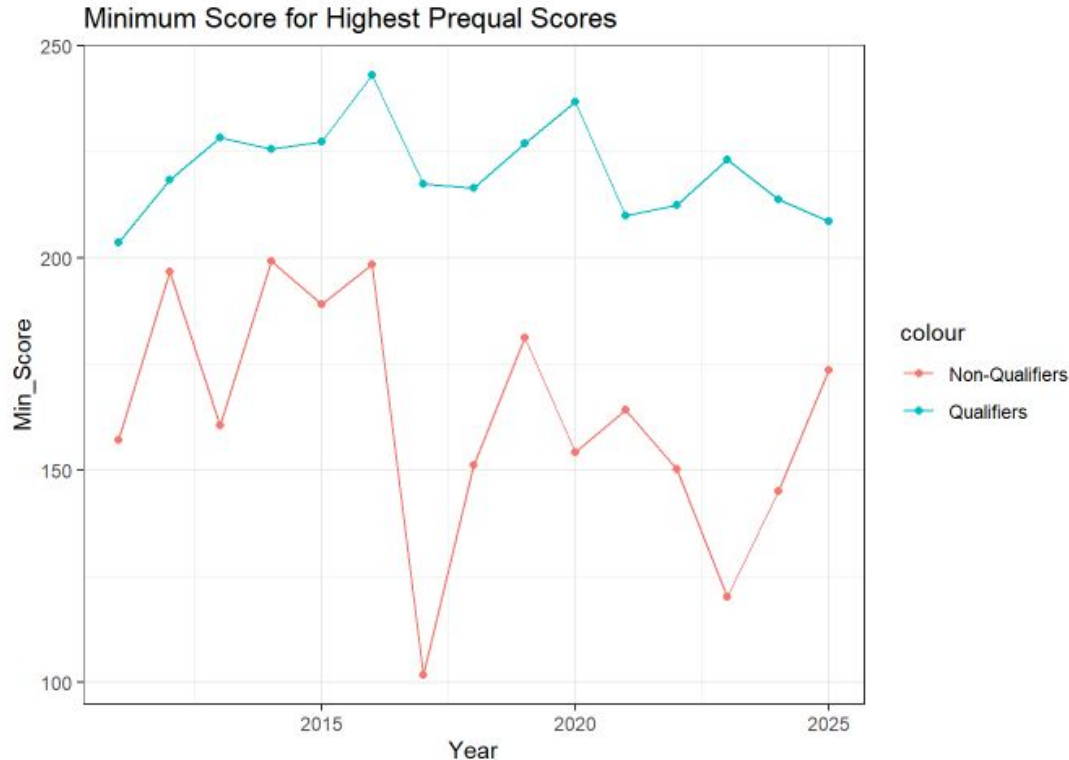


Exploring Potential Predictors: International

- International divers have higher average scores (224.5621 vs. 248.1875).
- However, this variable did not significantly improve model performance, so it was excluded.



What timeline should we consider?



- Year 2020 was most likely not a usual season due to COVID→ not included
- NCAA diving qualification standards and competition formats can shift over time, so older data may not reflect the more recent competitive landscape.
- We have decided to use all years post-COVID, which is **2021, 2022, 2023, 2024, 2025** in our analysis

Statistical Analysis

What type of model is appropriate?

- Essentially a classification problem (qualified or not)
- **Logistic regression**
 - Allows us to give interpretable probability. Rather than predicting qualified or not, we can provide divers with a probability of qualifying given certain characteristics/scores, which is much more actionable
- Other methods exist (LDA, RF, SVM, etc)
 - Were not considered in our analysis due to our priority of **striking a balance between performance and interpretation**

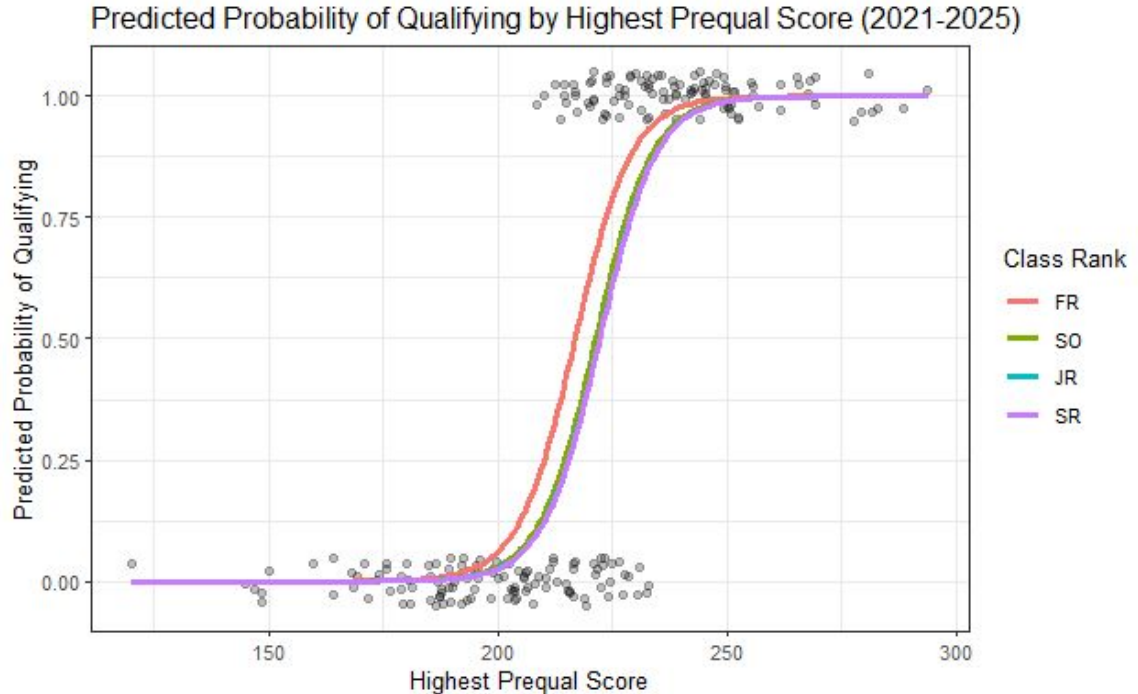
Initial Pass of Fitting Logistic Regression

- fit1: qualify ~ highest_prequal (AIC: 119.84)
- fit2: qualify ~ highest_prequal + class_rank (AIC: 123.56)

Let's take a closer look at the class rank predictor

Revisiting the Class Rank Predictor

- After filtering our dataset to only include recent, post-Covid years (2021-2025) fitting **fit2**
- Logistic curve for class rank shows that Freshmen score significantly lower than other classes
- **Indicator variable Freshman or not a Freshman** may be predictive of qualifying.



Logistic Regression Analysis

- Logistic Regression Models that were fit:
 - fit1: qualify ~ highest_prequal (AIC: 119.84)
 - fit2: qualify ~ highest_prequal + freshman (AIC: 119.62)
 - fit3: qualify ~ highest_prequal * freshman (AIC: 120.62)
 - fit4: qualify ~ highest_prequal + freshman + international (AIC: 119.78)
- Best fitted model: fit2
 - fit2: qualify ~ highest_prequal + freshman**
 - Why? Lowest AIC of 119.62, significant p-values of predictors while capturing the freshman effect.

$$\log(p/(1-p)) = \text{beta}_0 + \text{beta}_1 * (\text{highest prequal score}) + \text{beta}_2 * (\text{freshman indicator})$$

Cross Validation

- To evaluate our model's performance, we used a leave-one-out cross-validation by looping through each of the five years
- We used a cutoff (probability of qualifying) of 0.9, which means we predicted a diver qualifies if the model is at least 90% confident
- We chose a higher threshold so it is meaningful to the divers

Example:





Cross Validation Results

Baseline: Qualify ~ highest_prequal

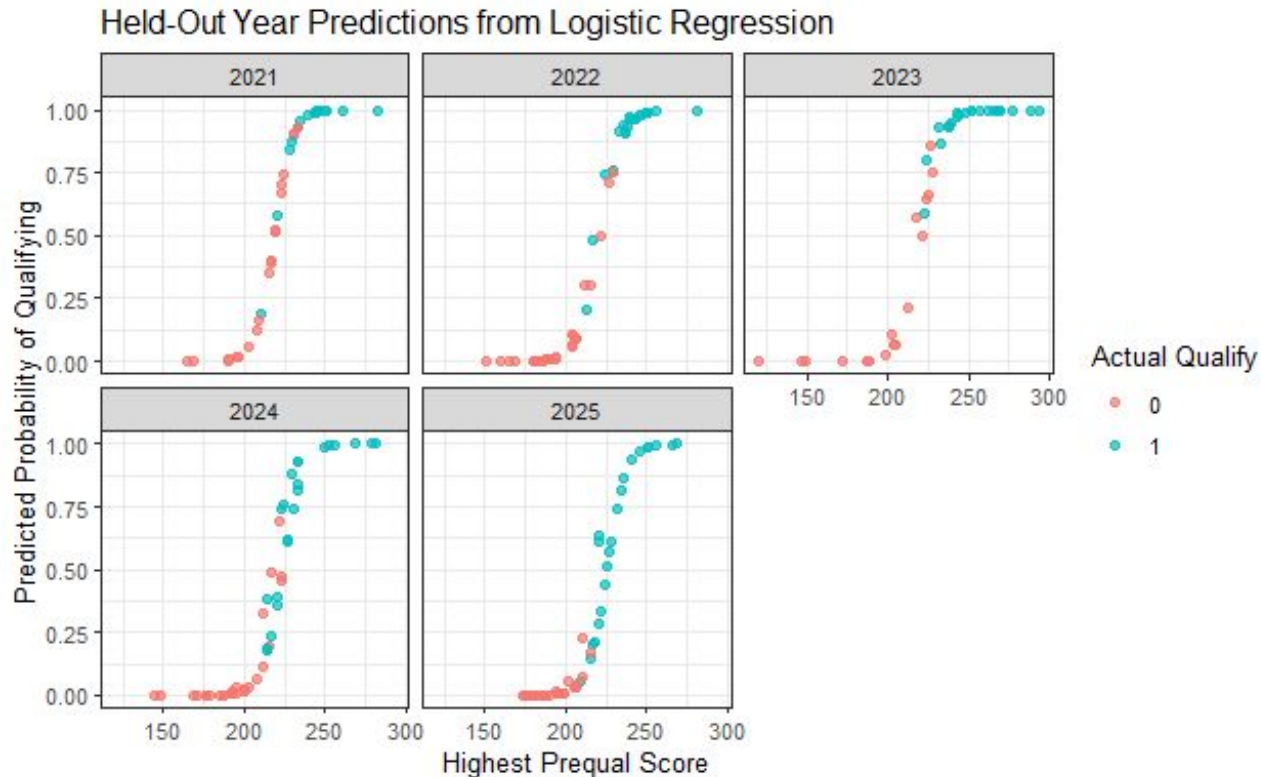
Classification Metrics				Confusion Matrix				
Accuracy	Sensitivity	Specificity	AUC	TP	TN	FP	FN	Year
78.4%	75.0%	81.0%	0.917	12	17	4	4	2021
89.8%	77.3%	100.0%	0.985	17	27	0	5	2022
89.7%	81.8%	100.0%	0.979	18	17	0	4	2023
66.7%	27.3%	100.0%	0.958	6	26	0	16	2024
65.9%	31.8%	100.0%	0.992	7	22	0	15	2025

Qualify ~ highest_prequal + freshman

Classification Metrics				Confusion Matrix				
Accuracy	Sensitivity	Specificity	AUC	TP	TN	FP	FN	Year
78.4%	75.0%	81.0%	0.917	12	17	4	4	2021
91.8%	81.8%	100.0%	0.985	18	27	0	4	2022
92.3%	86.4%	100.0%	0.987	19	17	0	3	2023
70.8%	36.4%	100.0%	0.946	8	26	0	14	2024
65.9%	31.8%	100.0%	0.983	7	22	0	15	2025

Recall: Sensitivity = true positive rate (correctly identifying qualifiers);
Specificity = true negative rate (correctly identifying non qualifiers)

Predicted Probability of qualifying for qualify ~ highest_prequal + freshman by heldout year



Overall Method Results for qualify ~ highest_prequal + freshman

accuracy <dbl>	sensitivity <dbl>	specificity <dbl>	auc <dbl>	tp <int>	tn <int>	fp <int>	fn <int>
0.797235	0.6153846	0.9646018	0.9423928	64	109	4	40

The model performs very well overall, especially with high specificity, meaning it correctly identifies non-qualifiers.

Discussion and Limitations

Conclusions

- Using the **last five years** in the dataset, we have found that the best model (AIC: 119.62) using a **logistic regression** is using the predictors of **highest prequalification score** and the **freshman indicator variable**
- Limitations
 - Small sample sizes per year (only 20-30 athletes per year)
 - Single program focus → Clarion University was dominant so a model trained heavily on one program may not do well in predicting qualification across all NCAA programs

Thank you!

Any questions?

Additional Figures

Unselected Model Evaluations

Fitted Results for qualify ~ highest_prequal

```
Call:
glm(formula = qualify ~ highest_prequal, family = binomial, data = model_data_new)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   -34.73294    5.23306  -6.637 3.20e-11 ***
highest_prequal  0.15747    0.02365   6.658 2.77e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 300.45  on 216  degrees of freedom
Residual deviance: 115.84  on 215  degrees of freedom
AIC: 119.84

Number of Fisher Scoring iterations: 7
```

Fitted Results for qualify ~ highest_prequal + freshman

SELECTED MODEL

```
Call:
glm(formula = qualify ~ highest_prequal + freshman, family = binomial,
     data = model_data_new)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-35.99598	5.46898	-6.582	4.65e-11	***
highest_prequal	0.16223	0.02456	6.605	3.99e-11	***
freshman	0.81329	0.55525	1.465	0.143	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 300.45 on 216 degrees of freedom
 Residual deviance: 113.62 on 214 degrees of freedom
 AIC: 119.62

Number of Fisher Scoring iterations: 7

Fitted Results for `qualify ~ highest_prequal * freshman`

```
Call:
glm(formula = qualify ~ highest_prequal * freshman, family = binomial,
     data = model_data_new)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   -33.44959    5.67671  -5.892 3.81e-09 ***
highest_prequal    0.15078    0.02550   5.912 3.37e-09 ***
freshman      -14.40050   16.79437  -0.857   0.391
highest_prequal:freshman  0.06969    0.07696   0.905   0.365
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 300.45  on 216  degrees of freedom
Residual deviance: 112.62  on 213  degrees of freedom
AIC: 120.62

Number of Fisher Scoring iterations: 7
```


Fitted Results for qualify ~ highest_prequal + freshman + international

```
Call:
glm(formula = qualify ~ highest_prequal + freshman + international,
     family = binomial, data = model_data_new)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-36.60066	5.62657	-6.505	7.77e-11	***
highest_prequal	0.16462	0.02525	6.520	7.03e-11	***
freshman	0.82277	0.55413	1.485	0.138	
international1	1.50685	1.14244	1.319	0.187	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 300.45 on 216 degrees of freedom
 Residual deviance: 111.78 on 213 degrees of freedom
 AIC: 119.78

Number of Fisher Scoring iterations: 7

We tried using all years without 2020, but our model did not perform nearly as well

```
Call:
glm(formula = qualify ~ highest_prequal, family = binomial, data = model_data_2020)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   -34.75657    3.17709  -10.94  <2e-16 ***
highest_prequal  0.15531    0.01415   10.97  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 786.39  on 568  degrees of freedom
Residual deviance: 301.36  on 567  degrees of freedom
AIC: 305.36

Number of Fisher Scoring iterations: 7
```

CV for highest_prequal*international wasn't as good

accuracy <dbl>	sensitivity <dbl>	specificity <dbl>	auc <dbl>	tp <int>	tn <int>	fp <int>	fn <int>	held_out_year <int>
0.8378378	0.8750000	0.8095238	0.9255952	14	17	4	2	2021
0.8979592	0.7727273	1.0000000	0.9865320	17	27	0	5	2022
0.8974359	0.8181818	1.0000000	0.9786096	18	17	0	4	2023
0.6666667	0.2727273	1.0000000	0.9580420	6	26	0	16	2024
0.6818182	0.3636364	1.0000000	0.9917355	8	22	0	14	2025

Prediction Results for qualify ~ highest_prequal

